

Game Theory-Assignment2

Xinyi Yang

January 20th 2026

Exercise 3

Consider the zero-sum matrix game

$$A = \begin{pmatrix} -1 & -3 & -3 & 1 \\ 0 & 1 & -2 & -1 \\ 2 & -2 & 0 & -1 \end{pmatrix},$$

where player 1 (the minimizer) selects rows and player 2 (the maximizer) selects columns. Player 1 moves first.

(1)

A rational policy for player 2 is given by the following conditional strategy:

- If player 1 selects row 1, player 2 selects column 4, leading to a reward of 1.
- If player 1 selects row 2, player 2 selects column 2, leading to a reward of 1.
- If player 1 selects row 3, player 2 selects column 1, leading to a reward of 2.

The corresponding rational response for player 1 is to select either row 1 or row 2.

(2)

Assume player 1 uses the policy of selecting row 1 or row 2. Under this policy, the maximum payoff achievable by player 2 is 1, and no unilateral deviation can improve this outcome. Hence, player 2 does not regret using the above conditional policy.

Conversely, assume player 2 uses the conditional policy described above. The resulting outcomes for player 1 are

$$1 \quad (\text{row 1}), \quad 1 \quad (\text{row 2}), \quad 2 \quad (\text{row 3}).$$

Since player 1 is the minimizer, no deviation can yield a payoff strictly smaller than 1. Therefore, player 1 does not regret selecting row 1 or row 2.

Exercise 4

Matrix A

$$A = \begin{pmatrix} 2 & -1 & 1 & -1 \\ -2 & -3 & 1 & -2 \\ -1 & -2 & -3 & -4 \\ 1 & -1 & 0 & -1 \end{pmatrix}$$

The security level for player 2 is

$$V(A) = -2,$$

and column 1 is a security policy for player 2.

The security level for player 1 is

$$V^-(A) = -1,$$

and row 3 is a security policy for player 1.

Since $V(A) \neq V^-(A)$, the game admits no pure saddle-point equilibrium.

Matrix B

$$B = \begin{pmatrix} 1 & 1 & -1 \\ -2 & 1 & -2 \\ 1 & 1 & -1 \end{pmatrix}$$

The security level for player 2 is

$$V(B) = 1,$$

and column 2 is a security policy for player 2.

The security level for player 1 is

$$V^-(B) = 1,$$

and rows 1, 2, and 3 are all security policies for player 1.

Therefore, the game has the following pure saddle-point equilibria:

$$(1, 2), \quad (2, 2), \quad (3, 2).$$

Exercise 5

(1) No pure saddle-point equilibrium

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

The row security level is

$$V^-(A) = \max_i \min_j A_{ij} = \max(-1, -1) = -1,$$

while the column security level is

$$V(A) = \min_j \max_i A_{ij} = \min(1, 1) = 1.$$

Since $V(A) \neq V^-(A)$, the game admits no pure saddle-point equilibrium.

(2) A single pure saddle-point equilibrium

$$B = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}$$

The row security level is

$$V^-(B) = \max(0, 1) = 1,$$

and row 2 is the unique security policy for player 1. The column security level is

$$V(B) = \min(1, 2) = 1,$$

and column 1 is the unique security policy for player 2. Therefore, the game has a unique pure saddle-point equilibrium at $(2, 1)$.

(3) Exactly two pure saddle-point equilibria

$$C = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

The row security level is

$$V^-(C) = \max(0, 1) = 1,$$

and row 2 is the unique security policy for player 1. The column security level is

$$V(C) = \min(1, 1) = 1,$$

and both columns are security policies for player 2. Hence, the game admits exactly two pure saddle-point equilibria.

(4) Exactly three pure saddle-point equilibria

Such a configuration is not possible in a 2×2 zero-sum game. Indeed, the number of pure saddle-point equilibria must equal

$$(\text{\#security policies of player 1}) \times (\text{\#security policies of player 2}),$$

which cannot be equal to 3.

(5) Exactly four pure saddle-point equilibria

$$D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

All rows and all columns are security policies for the two players, and the security levels satisfy

$$V(D) = V^-(D) = 0.$$

Therefore, all four entries of the matrix are pure saddle-point equilibria.

Homework #2**Student name :**

Exercise	weight	grade (0..4)
3.1	12.5%	4
3.2	12.5%	4
4A	12.5%	4
4B	12.5%	3
5.1	10%	4
5.2	10%	4
5.3	10%	4
5.4	10%	4
5.5	10%	4

Grades

- 4 – correct
- 3 – mostly correct
- 2 – half-and-half
- 1 – mostly incorrect
- 0 – not done